LMM Chapter 3 assignments:

Project 1: In this project, your task is to complete the R code to duplicate some results and figures in your textbook for Chapter 3 rat pup example, as well as compliment the R output (e.g., model summary, hypothesis test result and plots) from section 3.4.3. The data “rat_pup.dat” is available on Canvas in the “Files” folder.

a) Data summary (section 3.2.2): Create R code to duplicate the weight table on page 63, FIGURE 3.1 on page 64 and FIGURE 3.2 on page 65. Also attach the R output table and plots in your project report (as pdf). PS: the pattern of the table and plots may look a little different from the SAS output due to the software difference.
   Hint: you may find R functions summarize() and bwplot() are helpful.
b) Model analysis and hypothesis tests (section 3.3): By following the instructions in section 3.4.3.1 and using the lme() function in R, perform the model analysis and hypothesis tests described in your textbook (where 6 models are considered in total). Though the textbook instructs the code in detail, it doesn’t print the R output. In your project report, compliment all the missing model summary, hypothesis test results and plots in the process (for all the 6 models and 6 Hypothesis tests).
c) From the R output obtained in b), choose the final model based on AIC. Does it result in the same final model as in the textbook? How about BIC as the choosing standard?
d) Diagnostics for the Final model (section 3.10): Create R code to duplicate the FIGURE 3.6, FIGURE 3.7, FIGURE 3.8 and FIGURE 3.9. Also attach the R output plots in your project report.
   Hint: you may find R functions histogram(), qqnorm() and bwplot() are helpful.

Team Project #1

Project 2: In this project, your task is to create two-level LMM models for the data “classroom6.RData”. You need to finish data summary, model analysis, hypothesis tests to choose the final model, the interpretation of the final model and diagnostics for the final model. The data includes 1,190 students sampled from 38 classrooms. Below is the data variables descriptions:

- SEX: Indicator variable (0=boy,1=girl)
- MATHKIND: Student’s math score in the spring of 2022
- CLASSID: classroom ID number
- YEARSTEA: for the specific class, the teacher’s years of teaching experience
- MATHPREP: for the specific class, the teacher’s math preparation level, determined by the number of math content and methods courses taught.
- CHILDID: student ID number
- MATHGAIN: student’s gain in math achievement score from the spring 2022 to the spring of 2023
- MINORITY: indicator variable (0=non-minority student, 1=minority student)
- SES: student socioeconomic status

MATHGAIN is the dependent variable, and you need to determine which variables above have significant impact on MATHGAIN as well as to create a two-level LMM model for the data. The “classroom6.RData” is available on Canvas in the files folder. Please download it and load it into your Rstudio. Follow the following steps to complete the project report.

a) Data structure: for each of the variables including the dependent variable, determine whether it is level 1 variable or level 2 variable, fixed factor or random factor. Fulfill the table by checking the appropriate blank.

	SEX	MATH KIND	CLA SSID	YEAR STEA	MATH PREP	CHILD ID	MATH GAIN	MINO RITY	SES
Level 1									
Level 2									
Fixed factor							DP		
Random factor									

b) After determine the data structure, you need to give the data summary.
b.1) Give the level 1 descriptive statistics and the level 2 descriptive statistics.
b.2) Use boxplots to discover the possible relationship between the dependent variables and other variables. You can analyze it for the whole population and by classes. Record your R plots in the report and give comments on each of them. State the potentially important relationship between the dependent variables and the covariates if you find any.

c) Model analysis, hypothesis tests and choose the final model: by following the “top-down” modeling strategy similar to the rate pup example (FIGURE 3.3), create at least 5 models and perform appropriate hypothesis tests for each choice, then determine your final two-level LMM model. Show your process and work in your project report (e.g. model summary for each model, hypothesis test results etc.)
Hint: There are 38 classes in total and it will include too many parameters if we assume heterogeneous residual error variance for each of the class groups. So it may be better to create a new variable “YEARSTEA_LEVEL” in the dataset which classify the YEARSTEA into H,M, L level.

d) For the initial model, write down a general specification for an individual MATHGAIN observation similar to (3.1) on page 69 in the textbook.

e) Summarize your created models for comparison in a table like TABLE 3.3 on page 68 in the textbook.

f) Summarize your hypothesis tests in the comparison of each pair of models in a table like TABLE 3.4 on page 71 in the textbook.

g) Interpret the parameter estimates in your final model, including the fixed-effect parameter estimates and the covariance parameter estimates. (You may refer to section 3.7 page 120-121).
h) For your final model, give the associated equation (including parameter estimates as numbers and EBLUPs) to calculate the predicted value, both conditional and unconditional, like in section 3.9.
i) Give the residual diagnostics for your final model for the assumption of normality, and constant variance, both about conditional residuals and conditional studentized residuals. (Section 3.10.1.1-3.10.1.2)